Metropolitan University

Department of Computer Science & Engineering

Lab Report

Lab Task 3: Writing a Unique Essay Using LaTeX Commands

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Section: C

Course Code: CSE 429

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Abstract

This report provides a comprehensive overview of **Deep Learning (DL)**, a subfield of Machine Learning (ML) that leverages multi-layered artificial neural networks to automate feature extraction from raw data. We examine the structural foundation of neural networks, survey specialised architectures (CNNs, RNNs, Transformers), and discuss the computational requirements that underpin modern DL systems. Mathematical notation used throughout this document follows the standard LaTeX symbol conventions as referenced in the course material.

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1. Introduction

Deep Learning (DL) represents a sophisticated subfield of Machine Learning (ML) that leverages multi-layered computational models to extract high-level features from raw data. While traditional machine learning often requires manual feature engineering, Deep Learning automates this process through the use of **Artificial Neural Networks (ANNs)**.

By mimicking the hierarchical processing of the human brain, these systems have revolutionised how machines interpret visual, auditory, and textual information. Formally, a deep neural network defines a parametric function

$$f_{\boldsymbol{\theta}} : \mathcal{X} \longrightarrow \mathcal{Y}, \qquad \boldsymbol{\theta} \in \mathbb{R}^d, \tag{1}$$

where $\mathcal{X}$ is the input space, $\mathcal{Y}$ the output space, and $\boldsymbol{\theta}$ the learnable parameters (weights and biases).

2. Structural Foundation: Neural Networks

At the core of Deep Learning is the neural network architecture. These models consist of interconnected nodes (*neurons*) organised into distinct functional layers:

- **Input Layer** — Receives raw data (e.g. pixel values $\mathbf{x} \in \mathbb{R}^n$).
- **Hidden Layers** — Each layer ℓ applies an affine transformation followed by a non-linear activation σ:

$$\mathbf{h}^{(\ell)} = \sigma\Big(\mathbf{W}^{(\ell)}\,\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\Big), \quad \ell = 1, \ldots, L.$$

- **Output Layer** — Provides the final prediction or classification $\hat{y} = f_{\boldsymbol{\theta}}(\mathbf{x})$.

2.1. Backpropagation and Gradient Descent

Parameters are updated by minimising an empirical risk (loss) function $\mathcal{L}(\boldsymbol{\theta})$ via **stochastic gradient descent (SGD)**:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\,\nabla_{\boldsymbol{\theta}}\,\mathcal{L}(\boldsymbol{\theta}_t), \tag{2}$$

where $\eta > 0$ is the learning rate. Gradients are propagated backwards through the network using the *chain rule*:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(\ell)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(\ell)}} \cdot \frac{\partial \mathbf{h}^{(\ell)}}{\partial \mathbf{W}^{(\ell)}}. \tag{3}$$

3. Specialised Architectures and Applications

Deep Learning utilises specialised architectures tailored for specific data types. Table 1 summarises the most prominent families.

Table 1: Overview of major Deep Learning architectures.

Model Type	Primary Application	Key Example
CNN	Image & Video Processing	Facial recognition, medical imaging
RNN	Sequential Data	Speech-to-text, time-series forecasting
Transformer	Natural Language Processing	LLMs, real-time translation
GAN	Generative Modelling	Image synthesis, data augmentation

3.1. Convolutional Neural Networks (CNNs)

A CNN applies learnable filters $\mathbf{K} \in \mathbb{R}^{k \times k}$ to an input feature map via discrete convolution:

$$(I * K)(i, j) = \sum_m \sum_n I(i + m,\ j + n) \cdot K(m, n). \tag{4}$$

3.2. Recurrent Neural Networks (RNNs) and Transformers

RNNs maintain a hidden state $\mathbf{h}_t$ across time steps: $\mathbf{h}_t = \tanh(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t)$. Transformers replace recurrence with a *self-attention* mechanism:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{5}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are query, key, and value matrices, and d_k is the key dimension.

4. Computational Requirements and Scalability

The efficacy of Deep Learning is inherently tied to two factors: **Big Data** and **Computational Power**. Unlike traditional algorithms that may plateau, DL models satisfy an approximate *power law* between data volume N and generalisation error ε:

$$\varepsilon(N) \ \approx\ C \cdot N^{-\alpha}, \qquad \alpha > 0. \tag{6}$$

To handle the immense mathematical operations required for training, researchers utilise:

- **GPUs** — Massively parallel floating-point units.
- **TPUs** — Custom ASICs optimised for tensor operations ($\sum, \prod, \int$).

These accelerators reduce training times from months to hours.

5. Mathematical Symbol Reference

The following symbols, drawn directly from the course reference sheet, appear throughout this report:

Symbol	Meaning	Symbol	Meaning
∂	Partial derivative	∇	Gradient (nabla)
∞	Infinity	$\sum$	Summation
$\prod$	Product	$\int$	Integral
$\forall$	For all	$\exists$	There exists
$\in$	Element of	$\subseteq$	Subset
$\leq$	Less than or equal	$\geq$	Greater than or equal
$\approx$	Approximately equal	$\equiv$	Equivalent
$\mathbb{R}$	Real numbers	$\mathbb{N}$	Natural numbers
α, β	Greek: alpha, beta	θ, σ	Greek: theta, sigma
λ, μ	Greek: lambda, mu	ϵ, δ	Greek: epsilon, delta

6. Conclusion

Deep Learning has transitioned from a theoretical concept to the backbone of global digital infrastructure. Its ability to solve non-linear, complex problems—characterised mathematically by the universal approximation theorem—has made it indispensable in sectors ranging from precision medicine to algorithmic finance.

As we move toward a data-centric future, the continued evolution of Deep Learning will remain the primary catalyst for achieving truly autonomous and intelligent systems.

References

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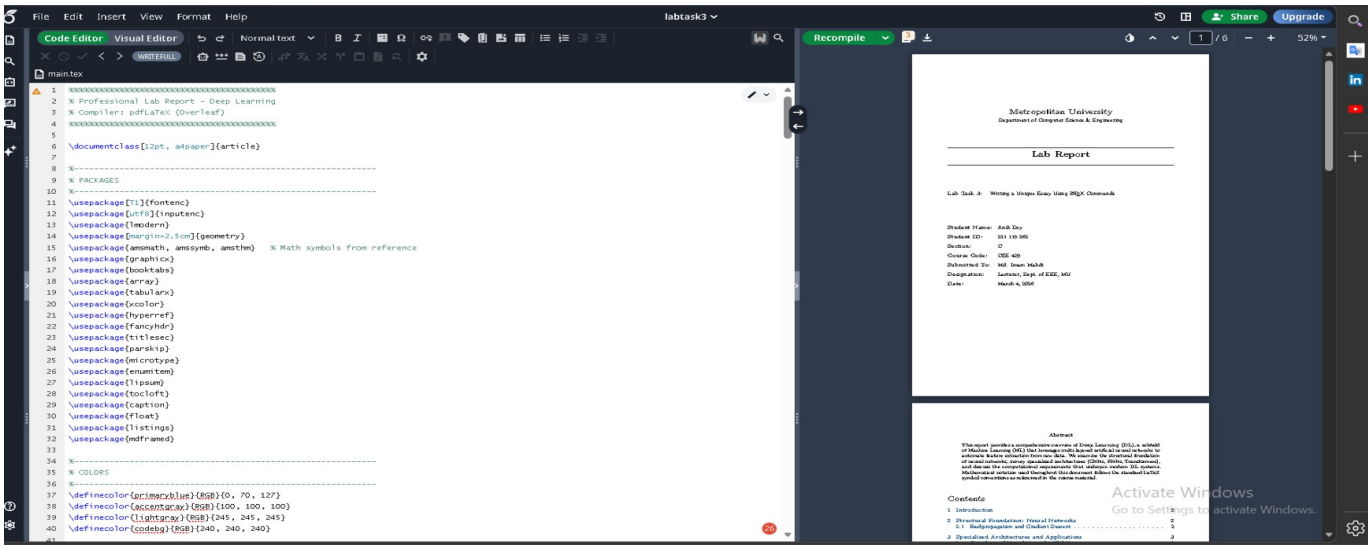
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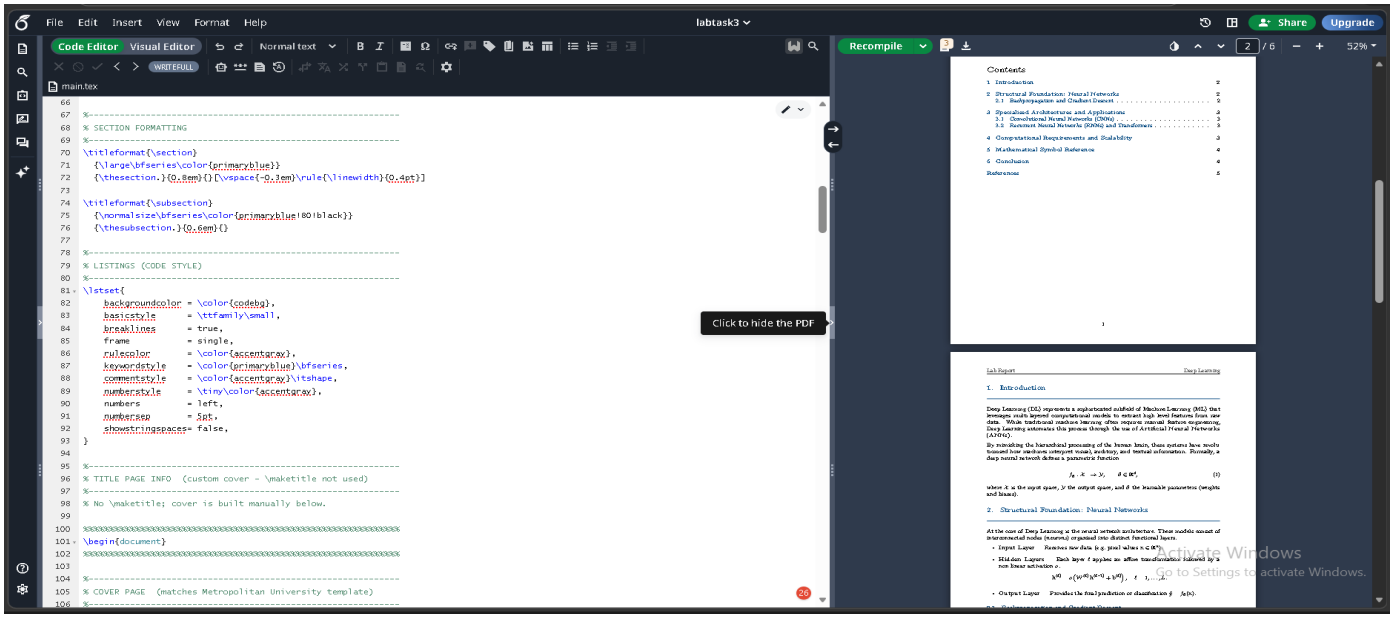
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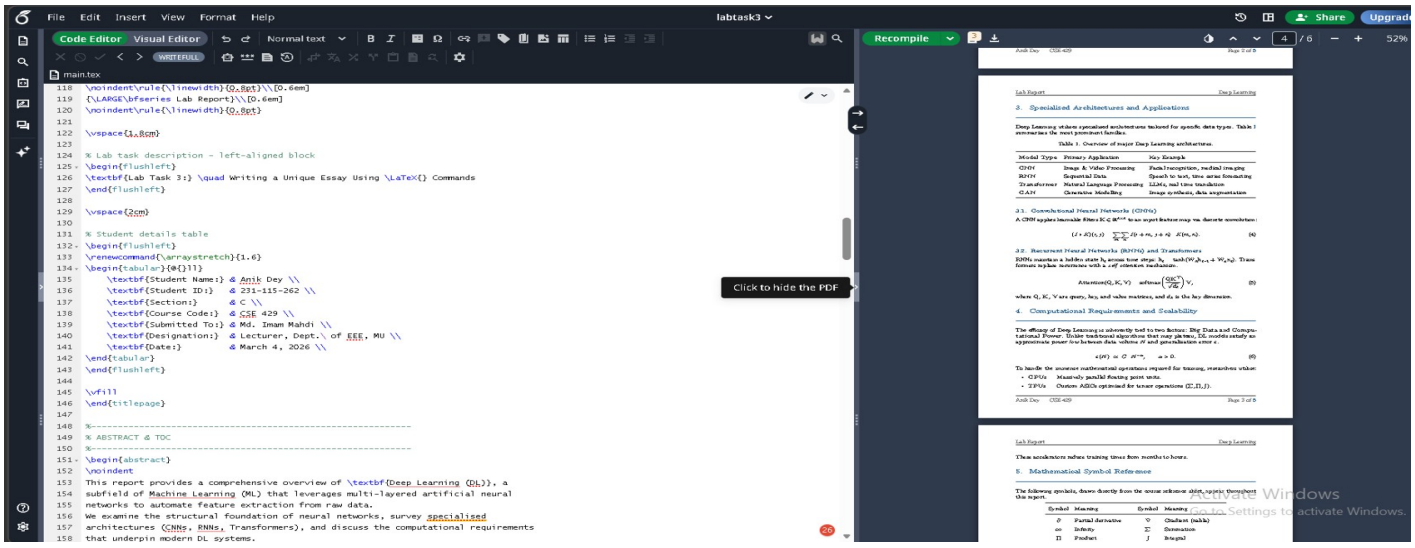
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3. Specialized Architectures and Applications
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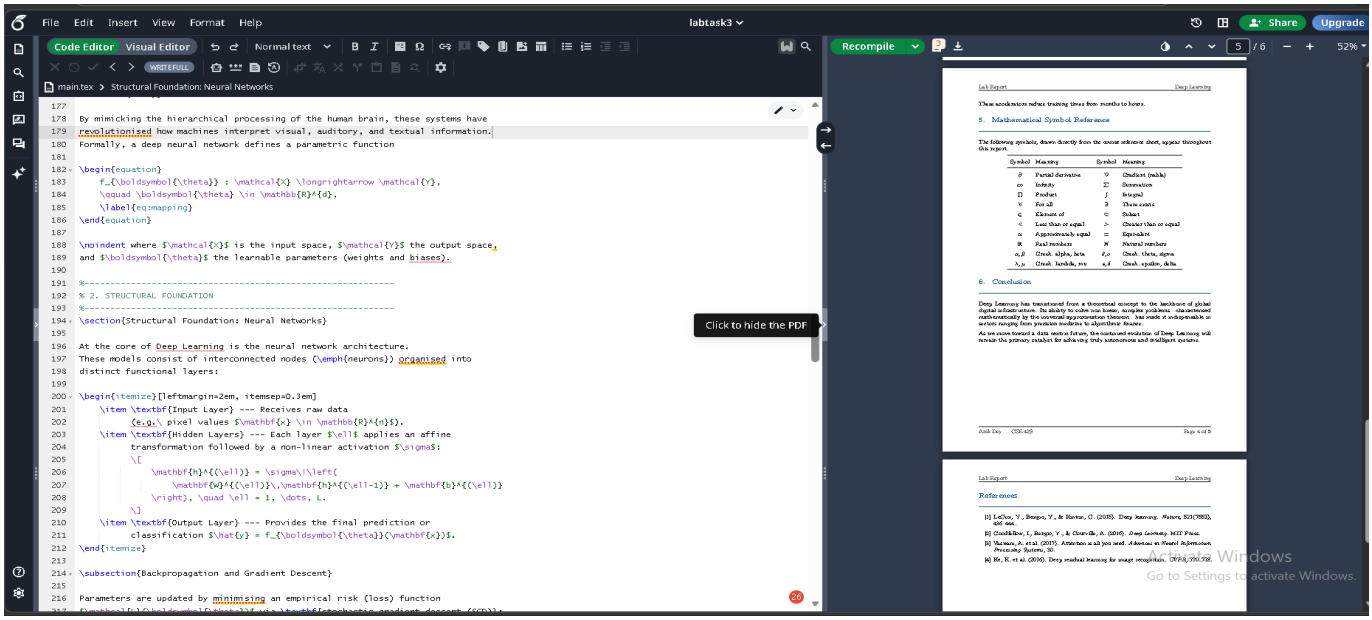

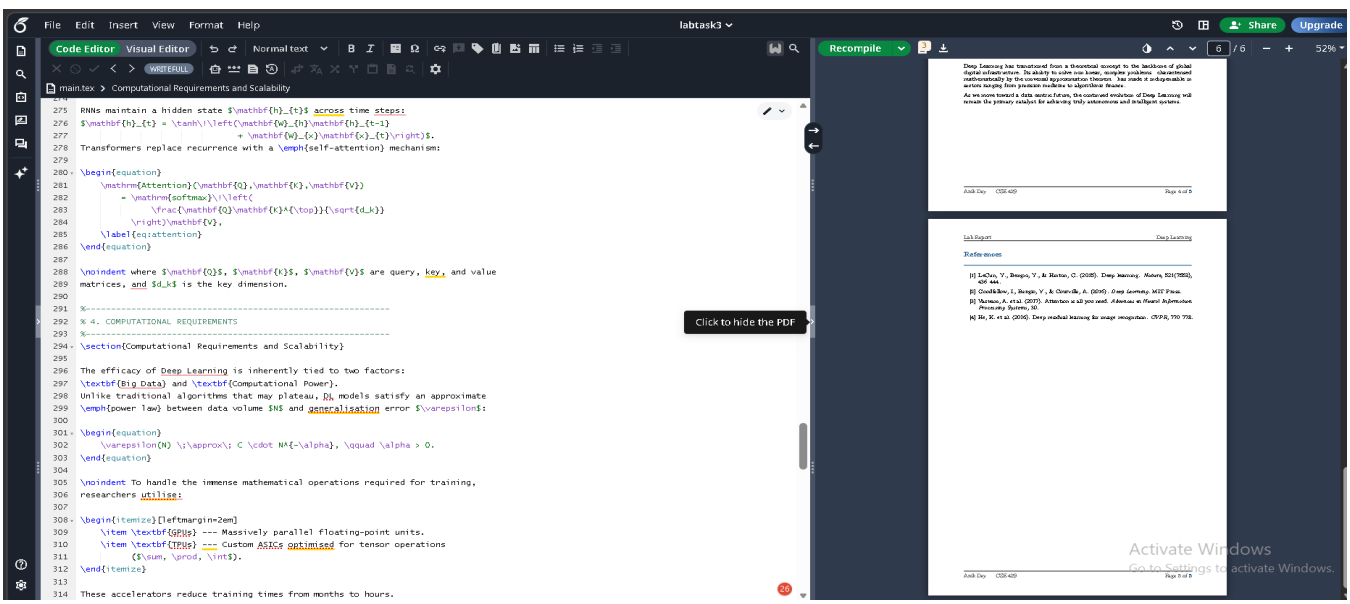

Description of Commands Used :

In this lab task, various LaTeX commands were used to prepare the document in a structured and professional format.

The **`\documentclass{}`** command was used to define the type of document. The **`\title{}`**, **`\author{}`**, and **`\date{}`** commands were used to provide basic information about the report. The `\maketitle` command was used to generate the title section automatically.

Sectioning commands such as **`\section{}`** and **`\subsection{}`** were used to organize the content into clear parts. This helped in maintaining proper structure and readability.Mathematical

expressions were written using math mode with **`$...$`** for inline equations and **`\[...\]`** or **`\begin{equation}...\end{equation}`** for displayed equations. Commands like **`\sum`**, **`\int`**, **`\partial`**, **`\nabla`**, and Greek letters such as **`\alpha`**, **`\theta`**, and **`\sigma`** were used to represent mathematical symbols properly.

The **`\begin{table}`** and **`\end{table}`** environment was used to create tables. The **`\begin{itemize}`** environment was used for bullet points, and **`\begin{enumerate}`** was used for numbered lists.

The **`\tableofcontents`** command was used to generate the table of contents automatically. Page numbering and formatting were handled using standard LaTeX formatting commands.

These commands helped create a clean, organized, and professional academic report.

Other Features Used

Several additional features were used to improve the quality of the report:

- Automatic page numbering
- Proper alignment of equations
- Professional table formatting
- Mathematical symbol formatting
- Structured headings and subheadings
- Reference section formatting

These features made the document more readable and academically strong. The use of mathematical notation improved clarity and precision in explaining Deep Learning concepts.

Challenges Faced

During the preparation of this report, several challenges were faced.

One major challenge was writing complex mathematical equations correctly in LaTeX format. Small syntax mistakes sometimes caused errors in compilation.

Another challenge was formatting tables and aligning equations properly. Ensuring that all sections were structured correctly also required careful attention.

Understanding and correctly applying mathematical symbols such as gradients, partial derivatives, and summation notation required extra focus.

Managing references in proper academic format was also slightly challenging at first.

However, through practice and careful revision, these challenges were successfully solved. This task improved my understanding of LaTeX formatting and professional report writing skills.